# project1 — Requirements

Household Grocery Food Log is a cloud-hosted web application for a single household where members log grocery shopping trips, track individual food items, mark items as consumed, and audit all activity. It is self-contained with its own username/password authentication and a two-level role system (Admin, Member).

GOALS
- Give every household member one simple place to log grocery purchases with item details.
- Keep an accurate live inventory of food that is still available (not yet consumed).
- Provide a full audit trail of who did what and when.
- Let the household Admin manage members and roles.

USER ROLES
- Admin: Full access. Manages household members (add, deactivate, promote/demote), can add/edit/remove food logs, sees the full activity log, manages the app.
- Member: Can add grocery shopping entries, mark food as consumed, and view the activity log. Cannot manage users or change roles.

EPICS & FEATURES
Epic 1: Authentication & Access
1.1 Login — simple username/password login for household members.
1.2 Role-based access — Admin vs Member permissions enforced across screens and API.
- As a member, I want to log in with my username and password, so that only my household can access our data.
Epic 2: Grocery Logging
2.1 Add Grocery Shopping — form to log a shopping trip (date, optional store name) with multiple food items (name, quantity, unit, price, purchase date).
2.2 Dashboard overview — recently added groceries, current inventory summary, quick stats (total items, total spent this month), quick-add button.
- As a member, I want to add multiple items in one shopping trip, so that logging a whole grocery run is fast.
- As a member, I want to see what food we still have, so that I know what to buy next.
Epic 3: Inventory & Consumption
3.1 Food Log List — searchable/filterable table of all logged items (name, quantity, unit, price, purchase date, added by, status) with filters for status, date range, and member.
3.2 Mark as Consumed — one-click action that flags an item consumed, records who and when, and removes it from active inventory.
- As a member, I want to mark food as consumed, so that our inventory stays accurate.
Epic 4: Activity & Audit
4.1 Activity Log — chronological list of all actions (who did what and when), filterable by user and

action type; visible to all members.
- As a member, I want to see who added or consumed food, so that everyone in the household stays accountable.
Epic 5: Household Administration
5.1 User Management (Admin only) — list of members with roles; add new members, promote/demote between Admin and Member, deactivate members.
- As an admin, I want to manage household members and their roles, so that access stays under control.

BUSINESS RULES & CONSTRAINTS
- Only Admin can add/remove users and change roles.
- All members (Admin + Member) can add grocery shopping and mark items as consumed.
- An item can only be marked as consumed once (no double-consumption).
- Consumed items remain in the database for history/audit but are excluded from the active inventory view by default.
- The Activity Log is read-only for Members; Admin has full visibility.
- A user cannot deactivate themselves.
- Passwords must be hashed with bcrypt (never stored in plain text).
- Every meaningful action (add shopping, consume item, user management) creates an ActivityLog entry.

NON-FUNCTIONAL REQUIREMENTS
- Cloud-hosted web app: Node.js + Express API, MySQL database, Bootstrap responsive UI.
- Single household deployment; no external integrations at this stage.
- Responsive UI usable on phones while shopping.